

MULTIFLEX INTELLIGENT TERMINAL

USER'S GUIDE

* The 10 MHz xtal
has been replaced
with a 13 MHz xtal.

COPYRIGHT NOTICE

(c) COPYRIGHT, 1982 by MULTIFLEX TECHNOLOGY INC. All rights reserved worldwide. No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any human or computer language, in any form or by any means, electronic, mechanical, magnetic, optical, chemical, manual, or otherwise, without the express written permission of MULTIFLEX TECHNOLOGY INC., 319 COLLEGE ST., TORONTO, ONT., M5T 1S2.

The MULTIFLEX Intellegent Terminal, including all the hardware, software and documentation, is copyrighted under Canadian copyright laws by MULTIFLEX TECHNOLOGY INC. It is against the law to copy or in any way reproduce the MULTIFLEX Intellegent Terminal or any part of it, including any of the hardware units and any of the software on any medium.

INTRODUCTION

The MULTIFLEX Intelligent Terminal is an interactive display terminal which can be used to enter, display, transmit and receive information to or from a computer, modem, printer, remote terminal, or other auxiliary device. It is configured to operate in local or remote mode, and is easily reconfigurable to emulate other existing terminals, or customized for several applications.

The terminal is a single board divided into two independent sections. The keyboard section consists of the key matrix with 56 keys, and an ASCII encoder. It is connected to the terminal and video control section which uses a Z-80 microprocessor to store and display the characters, and to control the serial interface.

Local mode allows stand-alone operation. Data entered through the keyboard is displayed immediately on the screen. Up to 100 lines of 80 characters each can be entered into the terminal, with the screen forming a 24-line window into this data. Editing commands allow this window to be moved to examine and modify any part of the stored data. The transmit command can then be used to send the stored data over the serial interface to its destination. This allows the entry and editing of over 4 screen pages without tying up an expensive computer or modem connection. With a printer connected, the terminal becomes a complete small-scale word-processing system.

Remote mode is the standard on-line terminal operation. Characters entered through the keyboard are sent out over the serial interface without being displayed or interpreted. The host computer echoes these characters back, together with its own data to be displayed on the 24-line by 80-character screen.

ASSEMBLY INSTRUCTIONS

- [1] Check all parts received in the box containing your kit against the parts list. Report any missing parts to your MULTIFLEX distributor as soon as possible so that they may be sent to you. Notice that there are two separate parts lists; one for the terminal itself and one for the keyboard encoder.
- [2] Place the board in front of you. Note that the side on which the components go has the white silk-screened patterns on it. The white line all the way across the board indicates the division between the terminal section and the keyboard and encoder sections.
- [3] Install all the .1 μ F bypass capacitors in the appropriate places as indicated on the board by an oval with no part number associated with it.
- [4] Install the rest of the capacitors, as indicated by the part numbers on the parts list and the circuit board. Note the polarity on the tantalum and electrolytic caps. If they are installed backwards, this could destroy the capacitor, and damage the board.
- [5] Install the resistors in the proper places.
- [6] Install the Single Inline resistor packs (SIPs), making sure that pin 1 on the pack, marked by a dot or a line, is facing the same direction as marked on the layout diagram.
- [7] Install the transistors and diodes in the appropriate places, with the polarity of the diodes corresponding to the diagram.
- [8] Install all the IC sockets on the board.
- [9] Install all the connectors on the board as indicated by the silk-screen and the parts list.
- [10] Install all the IC's on the board. Insure that they are inserted in the correct direction, since incorrect insertion can lead to damage to the chips.
- [11] Set up the various jumpers as described in the SYSTEM HARDWARE description.
- [12] Mount the keyboard on the board, and solder all the keyswitch connections.
- [13] Either install the optional power supply circuit or attach an appropriate power supply to the board.

OPERATING INSTRUCTIONS

With the terminal completely assembled, connected to an appropriate power supply, and connected to a video monitor (CRT), applying power should cause it to initialize itself and display a blinking block cursor at the upper left of the screen. On power-up the terminal is automatically initialized to the set of default characteristics; it is set to local mode and can be used immediately.

The keyboard produces the complete set of 128 ASCII character codes. The 95 displayable characters are stored and displayed on the screen, and the 32 control codes are interpreted as commands to control the operation of the terminal. The control codes are produced by holding down the control (CTRL) key while typing the appropriate key; they are indicated here by preceding the key symbol by a "^". Holding a key down longer than about half a second causes it to start repeating automatically at the selected rate. Four keys do not produce any codes. They are the two SHIFT keys, the CTRL key, and the CAPS LOCK key.

The control key commands are executed when they are received. In the local mode, this is from the keyboard; in the remote mode this is from the serial interface.

One set of control key commands applies to both modes:

DEL - Not a control code, this separate key causes the character before (to the left of) the cursor to be deleted and replaced by the cursor; this forms a deleting backspace. With repeated use, the DEL key will go past the beginning of the line and continue deleting back from the end of the preceding line. Continuing past the beginning of the first line causes the screen to scroll down one line. In local mode it will not go past the beginning of the first line (on the first page); in remote mode it will continue past the beginning of the first line, onto the blank line just scrolled down.

BS - Also the ^H key, this is an ordinary backspace, moving the cursor one position left, without affecting the characters it passes over. With repeated use, the BS key will go past the beginning of the line and continue back from the end of the preceding line. Continuing past the beginning of the first line causes the screen to scroll down one line. In local mode it will not go past the beginning of the first line (on the first page); in remote mode it will continue past the beginning of the first line, onto the blank line just scrolled down.

^L - Forward Space (FS), moves the cursor one position right, without affecting the characters it passes over. With repeated use, ^L will go past the end of the line

and continue from the beginning of the next line. Continuing past the end of the last line causes the screen to scroll up one line. In local mode it will not go past the end of the last line (on the last page); in remote mode it will continue past the end of the last line, onto the blank line just scrolled up.

^J - Line Feed (LF), moves the cursor down to the next line in the same column position. Continuing past the last line on the screen causes the screen to scroll up one line. In local mode, it will not go past the 100th line; in remote mode it will continue past the 24th line, onto the blank line just scrolled up.

^K - Up Line, moves the cursor up to the preceding line in the same column position. Continuing past the first line on the screen causes the screen to scroll down one line. In local mode it will not go past the first line on the first page; in remote mode it will continue past the first line, onto the first line just scrolled down.

RET - Carriage Return (CR), also **^M**. In remote mode, this moves the cursor back to the beginning of the current line. In the local mode, it also does an automatic line feed, thus moving the cursor to the beginning of the next line; continuing past the last line on the screen will cause the screen to scroll up one line. It will not go past the 100th line.

TAB - Also **^I**, **TAB** moves the cursor to the next position which is a multiple of 8 from the beginning of the line. With repeated use, the **TAB** will go past the end of the line and continue from the beginning of the next line.

^P - End of Line, causes the cursor to move to the first position past the last non-space character on the line.

^^ - Home, causes the cursor to move to the first position of the first line, and the first line to be displayed at the top of the screen.

^Q - Clear Screen, clears the screen, deleting all the stored characters, and moves the cursor to the HOME position. (clear all pages in remot mode)

^G - Bell, generates an audible tone.

^F - Start Attribute, causes all further characters to be displayed with the selected attribute.

^D - Stop Attribute, causes all further characters to be displayed with the attributes turned off. This is the default condition on power-up.

^[- Escape (ESC), start of an ESCAPE sequence command. If

followed by an "=", it indicates the start of a load cursor command to allow absolute positioning of the cursor anywhere on the screen. The "=" must be followed by two more characters, the first one specifying the column number, and the second one specifies the row number. The numerical values of the characters follow the standard ASCII sequence with the space character (20H) producing a value of 1.

^@ - Enter or Exit Setup Mode. This command is only executed from the keyboard. The first **^@** causes entry into setup mode, displaying a status line on the next line following the following the cursor. The status line displays the currently-selected characteristics of the terminal:

Mode: Local (default) or remote - key 'M'
Screen: Standard (default) or Reverse video -key 'R'
Attribute : Dim (default), Off, Reverse Video, or
Alternate character set. -key 'A'
Baud Rate: 75, 110, 150, 300 (default), 600, 1200,
1800, 2400, 4800, or 9600 -key 'B'
Data Bits: 8 (default) or 7 -key 'D'
Parity: Off (default), Odd or Even -key 'P'
Stop Bits: 1 (default) or 2 -key 'S'

The indicated keys are used to step through the values for the corresponding parameter. Another **^@** will exit to the selected major mode, with the cursor returned to its previous position and the status line blanked. All other keys are ignored while in the setup mode.
NOTE: Switching to Remote Mode should only be done with the cursor on the first page (lines 1 to 24).

The following set of control code commands are available only in the local mode:

^X - Scroll up, will scroll the screen up one line. It will not scroll past the 100th line.

^E - Scroll down, will scroll the screen down one line. It will not scroll past the 1st line.

^C - Next page, will scroll the screen up 20 lines. It will not scroll past the 100th line.

^R - Previous page, will scroll the screen down 20 lines. It will not scroll past the 1st line.

^T - Transmit Display, will start sending the entire display over the serial interface, starting with the first character of the first line. Trailing spaces on a line and blank lines will be transmitted as a CR/LF (carriage return, line feed) pair to save on transmission time. During transmission, the cursor will step from line to line, indicating each line being transmitted. At this time, nothing can be recieved over the serial interface and ^@ stops

all keys will be ignored, except for A which will cause the transmission to be aborted. When transmission is completed, the cursor will be on the last character on the 100th line, and reception from the serial interface will be enabled again.

NOTE: Any line which is to be transmitted must have a space character in the last column position.

SYSTEM DESCRIPTION

The Z80 handles communications and manages the display buffer, while the 6845 controls the CRT. The main memory (RAM) of the terminal consists of 4 6116 (2K x 8) static RAMs giving a total of 8 Kbytes. Access to this memory is interleaved between the Z80 and the 6845. The Z80 can access the display memory at any time without interfering with the display. The terminal control software is supplied in a 2732 (4K x 8) EPROM, plugged into the bottom part of the 28 pin socket U5 (pin 1 of the 2732 into pin 3 of the socket). A 2764 EPROM can also be used to provide up to 8 Kbytes of user customized terminal controlled software.

The 8253 contains 3 independant 16-bit counters, dividing down from 2 MHz by software-programmable rates. Counter 0 is used for baud rate generation for the serial interface. Counter 1 is used as a timer to generate an optional NMI to interrupt the Z80, and counter 2 is used to provide a tone to the speaker connector JSPKR. Custom user software can be used to generate sequences of musical notes. The volume of the speaker is controlled by trimpot R1.

The 8251 provides an RS232C serial interface operating full duplex for communication with a host computer, modem, or directly with a printer, using the standard DB-25 connector at JPCI. A printer or other auxiliary device can be connected to the extension DB-25 connector at JPCIE, to allow simultaneous transfer of information to the display and a printer attached to this port. Jumpers J1-J6 can be set up as required for the printer. NOTE: These 2 connectors must be mounted on the bottom (solder side) of the board.

A power-on reset circuit is included, with provision for a reset push button connected to J9.

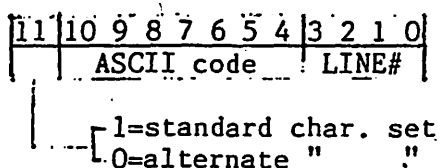
The supplied software operates both the keyboard and the serial interface receiver in interrupt mode. This requires the insertion of jumper J13 and J14 for the serial interface interrupt enable and the keyboard interrupt enable respectively. As an option for user software, jumper J11 is the programmable counter 1 (8253) timer interrupt enable, and jumper J12 is the CRT vertical sync (VSYNC) interrupt enable. Both of these are currently not used, but are available to generate an NMI interrupt to the Z80 if they are required for a special application.

The display's character generator is in a 2732 EPROM, providing 256 character symbols. This EPROM is organized in 2 sections: standard character set (located between 800H and FFFH) and alternate character set (located between 000H and 7FFH). Each ASCII code has 16 bytes for the scan lines of the character, as shown here:

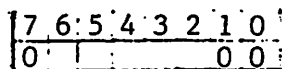
C60H 00000000
 C61H 01111100
 C62H 01000000
 C63H 01000000
 C64H 01111000
 C65H 01000000
 C66H 01000000
 C67H 01000000
 C68H 00000000
 C69H 00000000
 C6AH 00000000
 C6BH 00000000
 C6CH 00000000
 C6DH 00000000
 C6EH 00000000
 C6FH 00000000

WOULD DISPLAY AN "F" ON
THE SCREEN.

The EPROM addressing is:



The bits in each byte are:



1st dot of char:
in a 5x7 matrix

Last dot of char

There are 4 possible software selections of attributes: Dim, Reverse Video, Alternate Character Set or none of the above (standard video). Only 1 of these 4 attributes may be in effect at any one time. Characters stored in the display memory consist of a 7 bit ASCII code. The 8th bit (the MSB) is used to turn on and off the particular attribute in effect at the time.

Another attribute, the screen attribute affects the entire screen; it selects normal or reverse video for the whole screen. These attributes are selected by writing to the Keyboard Control Register (I/O port 0DFH), together with a display-on bit which enables or disables the 6845.

I/O port 0DFH: BIT#	USAGE
0	Attribute (*)
1	Attribute (*)
2	Display ON
3	Screen reverse video
4	Unused
5	"
6	"
7	"

(*) ATTRIBUTES: 00 = None

01 = Dim
10 = Reverse video
11 = Alternate char. set

Data from the keyboard is read via the Keyboard Data Register (I/O port 06FH). The Keyboard Control Register, when read, supplies the Keyboard Ready bit and 7 other bits which can be set by the DIP switches at J10.

The cursor can be software programmed as a block or underline, solid or blinking, fast or slow.

The composite video out (J19) is for a 75 ohm coax cable, with intensity controlled by trimpot R30.

An optional RF modulator is provided for connection to the antenna terminals of a standard TV set, using connector J17. This requires changing the crystal Y2 and changing the 6845's initialization parameters in the software.

THE KEYBOARD

The keyboard section is a general, independant keyboard complete with encoder. It includes the auto-repeat enable jumper J12, the auto-repeat start delay trimpot R6, and the repeat rate trimpot R12. The jumpers J4, J6, J9, and J10 must be inserted. J4 and J8 select handshake operation (J5 and J9 select direct strobe operation, which is not used here) J10 selects an active-low ready strobe (J11 selects active-high). J6 selects an active-low acknowledge handshake (J9 selects active-high). Interlocked handshake operation is used to eliminate lost characters at high typing speed. Options available on the keyboard are:

- Keyboard can be separated from the main board and connected with a 16-line cable between connectors JAKB, for both signals and power.
- Another matrix key "k1".
- "REP" key -repeat function selected by jumper J1
 -alternate key code function selected by jumper J2
- Alternate key codes selected by separate external switch connected to jumper J3.
- Character Generator EPROM supplied contains ASCII codes as shown in the diagram, with the MSB set to 0; optionally, the MSB can be used to encode 256 keycodes.
- Keypads with upto 32 additional keys can be connected to the "EXT" connector, with complete encoding through the EPROM.

DETAILED HARDWARE DESCRIPTION

PAGE 1: The Z80 (U11) controls the operation of the terminal, generating addresses ZAO-ZA15, with data transfers on the bidirectional data bus, ZD0-ZD7. U4 decodes the EPROM, RAM and I/O enables. One of these I/O devices is the 6845 CRT controller. It generates the memory addresses MA0-MA12 for refreshing the CRT display, the character row addresses RA0-RA3, and the CRT sync. control signals. The rest consists of logic to generate the RESET signal, the interrupt signals for the Z80, and the control signals from the Z80 to operate the memory and I/O devices.

PAGE 2: The 2732 (or 2764) EPROM (U5) provides the operating software. The Z80. the 8253 (U10) contains 3 software-programmable 16-bit counters, dividing down the 2 MHz crystal oscillator. Counter 0 generates the baud rate for the serial interface. Counter 1 provides an optional timer interrupt to the Z80. Counter 2 generates the tone for the speaker, which also has a volume control (R50). The 8251 (U9) is the serial interface, together with the RS232C driver (U3) and receivers (U1, U2). JPCI is a DB25 connector for a computer, modem or direct printer, and JPCIE is an extension DB25 connector for a printer operating simultaneously with the terminal.

PAGE 3: The 4 6116s (U17-U20) provide 8K of RAM for the display and system memory. This memory is addressed by the multiplexers U14-U16 and controlled by the multiplexer U24 plus the additional logic, to allow interleaved access from the Z80 and the 6845, with no wait states required and without interfering with the display.

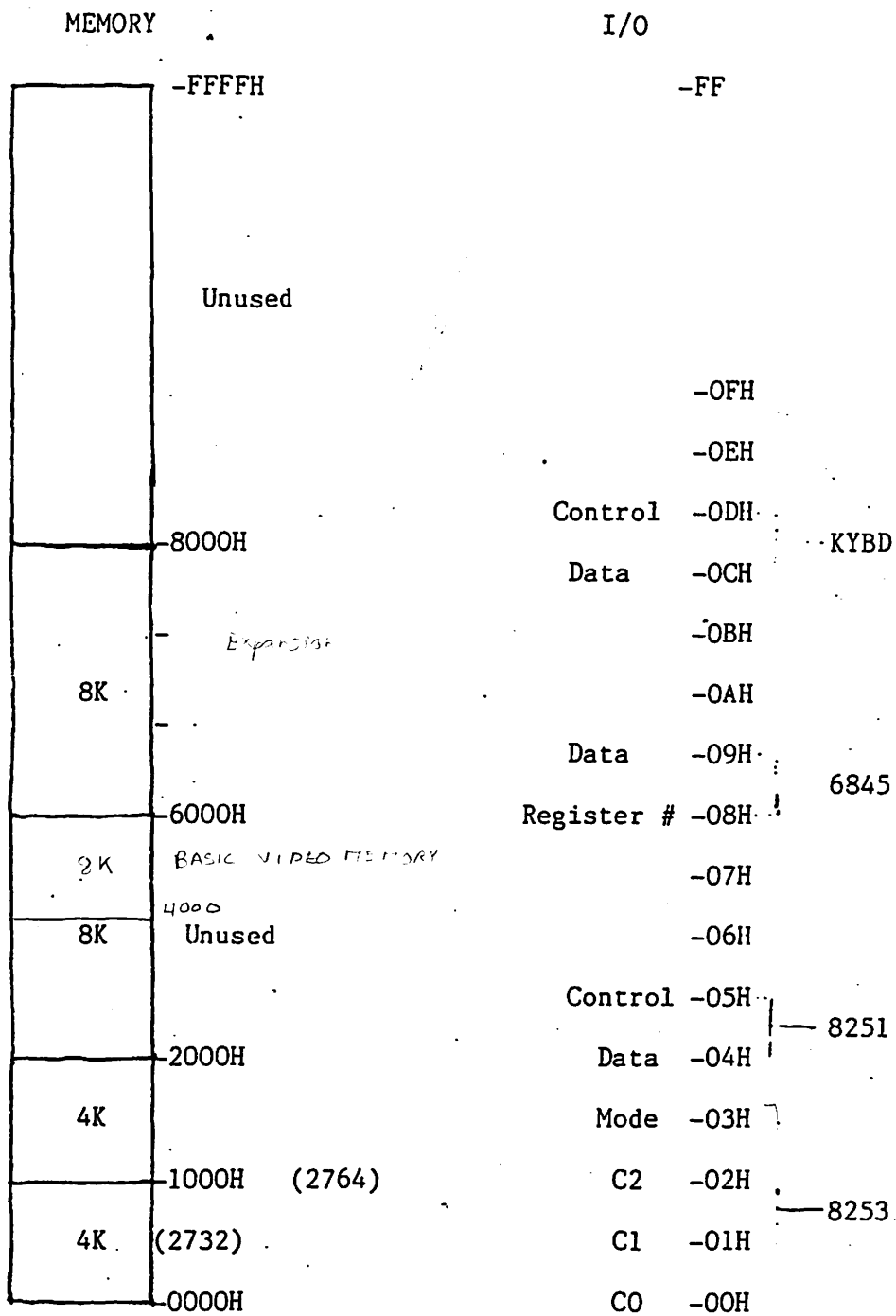
PAGE 4: The buffer U32 allows the Z80 to write data into this memory, and the latch U31 allows it to read from the memory. The data from this memory, as accessed by the 6845, is stored in register U33 and used to address the character generator 2732 EPROM (U34), together with the scanline-row address of the character. The data from this EPROM, consisting of the 5 dots on that row of the character is loaded into the shift register U40, and shifted out at the dot clock rate. This serial data is combined with the decoded attribute (U25) in U30, together with the cursor and blanking signals from the 6845. All these signals are delayed by U29 by 1 or 2 dots to synchronize with the character dot delays (in U33 and U34), and combined with the sync signals by U28, U38, and discrete logic. The counter U41 suitably shortens the VSYNC pulse. The output emitter follower Q3 drives the coaxial cable through the video connector J15.

PAGE 5: The oscillator U35 generates the 10 MHz dot clock, which is divided down by 6 (by the counter U36) to produce the character clock which forms the main time base for the terminal timing. The register U6 stores the selected attributes. The keyboard is connected through JKB and read by the Z80 through the buffer U8. Its ready status, together with 7 bits from a DIP switch are read by the Z80 through the buffer U7.

PAGE 6: Power to the terminal comes from a 14 volt center-tapped transformer and a 5 volt regulator VR1. The +10V and -10V section supplies the RS232C drivers. The bottom circuit is an RF modulator, with J17 for connection to the antenna terminals of a standard TV set.

KEYBOARD: Counter U4, driven by oscillator U12, scans the keyboard matrix rows through decoder U6. The output from the matrix columns is selected by multiplexer U5. One-shot U11 and flip-flop U8 provide the debounce delay, keeping the oscillator stopped as long as the key is held down. Flip-flop U8 latches the key-hit signal for the Connector Ready line, and is cleared by the handshake acknowledge signal from the Z80. U11 also provides the auto-repeat delay (adjusted by R6, and U12 generates the repeat rate (adjusted by R12). U3 latches the keyboard control keys and addresses the 2716 encoder EPROM (U2), together with the scanning counter. The data output from the EPROM consists of the corresponding ASCII code which is sent out over the connector through buffer U1.

MEMORY AND I/O MAP



JUNE, 1983

MULTIFLEX TERMINAL KIT

ADDENDUM

R10 is deleted

C13 and C14 are delete

R40, 41 and 42 are located behind connector JPCIE and are pull-down resistors to -12V line. (see schematic page 2 of 6).

100R trimpots maybe substituted for 50R.

Page 11 of the manual should read:

J4 and J9J5 and J8

(J7 selects.....

J7 on the Terminal Board is a 'direct serial accquission port' connected directly to JPCI.

The pinout is as follows:

J7	JPCI	SIGNAL
P1	N/C	N/A
P2	P20	DTR
P3	P2	TXD
P4	P5	CTS
P5	P3	RXD

J8 on the terminal Board is a direct voltage accquission point making available the +12V and GROUND SIGNALS.

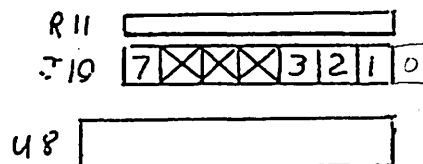
FOR A & I UNITS ONLY

Please note that the +5V, +12V, -12V, signals are present on JPCI; P9, P10, P11 respectively.

TMON 1.7 Additional Instructions

The following changes have been made in TMON 1.7

- A) J10 must be installed and will determine the default baud rate and local or remote mode on power up.
- B) [CTRL] Z will clear the screen.
- C) [CTRL] Ø will transmit a 320 ms break.
- D) A jump table and list of other important locations is available from Exceltronix upon request



Part	3	2	1
150 Baud	—	—	$\bar{X}$
300	—	$\bar{X}$	$\bar{X}$
600	—	$\bar{X}$	$\bar{X}$
1200	$\bar{X}$	X	$\bar{X}$
1800	$\bar{X}$	:	$\bar{X}$
2400	X	—	$\bar{X}$
4800	X	$\bar{X}$	—
9600	X	X	$\bar{X}$

(1)
Part 7 jumpered-remote mode
not jumpered-local mode
(0)

X = jumpered
— = not jumpered

Keybd control port: Read

- 0 - ACKOV from keybd (active high)
- 1 } Baud rate selection
- 2 } use xor/xor
- 3 } DTR high
- 4 - Not jumpered(1) - and xor if buffer has 122 chars, 0 = set DTR high only
- 5
- 6
- 7 - Remote / Local mode

Keybd Control port: Write

- 0 - Attribute 0
- 1 - Attribute 1
- 2 - Display on
- 3 - Reverse Video
- 4 } Attribute
- 5 } Attribute
- 6 } Attribute
- 7 } Attribute

.. TMON V1.7 Important Addresses

Address	Data	Condition
00A7	01	8 Data Bits
	00	7 Data Bits
00AC	00	No Parity
	01	Odd Parity
	03	Even Parity
00B1	00	1 Stop Bit
	01	2 Stop Bits
00B6	02	Inverse Video Attribute
	00	No Attribute
	01	Dim Attribute
	03	Alternate Character Set Attribute
00BB	00	Normal Video Screen
	01	Inverse Video Screen
010E	4x	Fast Blinking Cursor
	6x	Slow Blinking Cursor
	0x	Non-Blinking Cursor
	2x	Invisible Cursor
	x=0	Block Cursor
	x=8	Double Line Cursor
	x=9	Single Line Cursor
01FE	04	320 ms Break (This can be any value except 0, it is in increments of 80 ms.)

Reg10 (cursor start)

4 → 0 = Start line of cursor

6 5

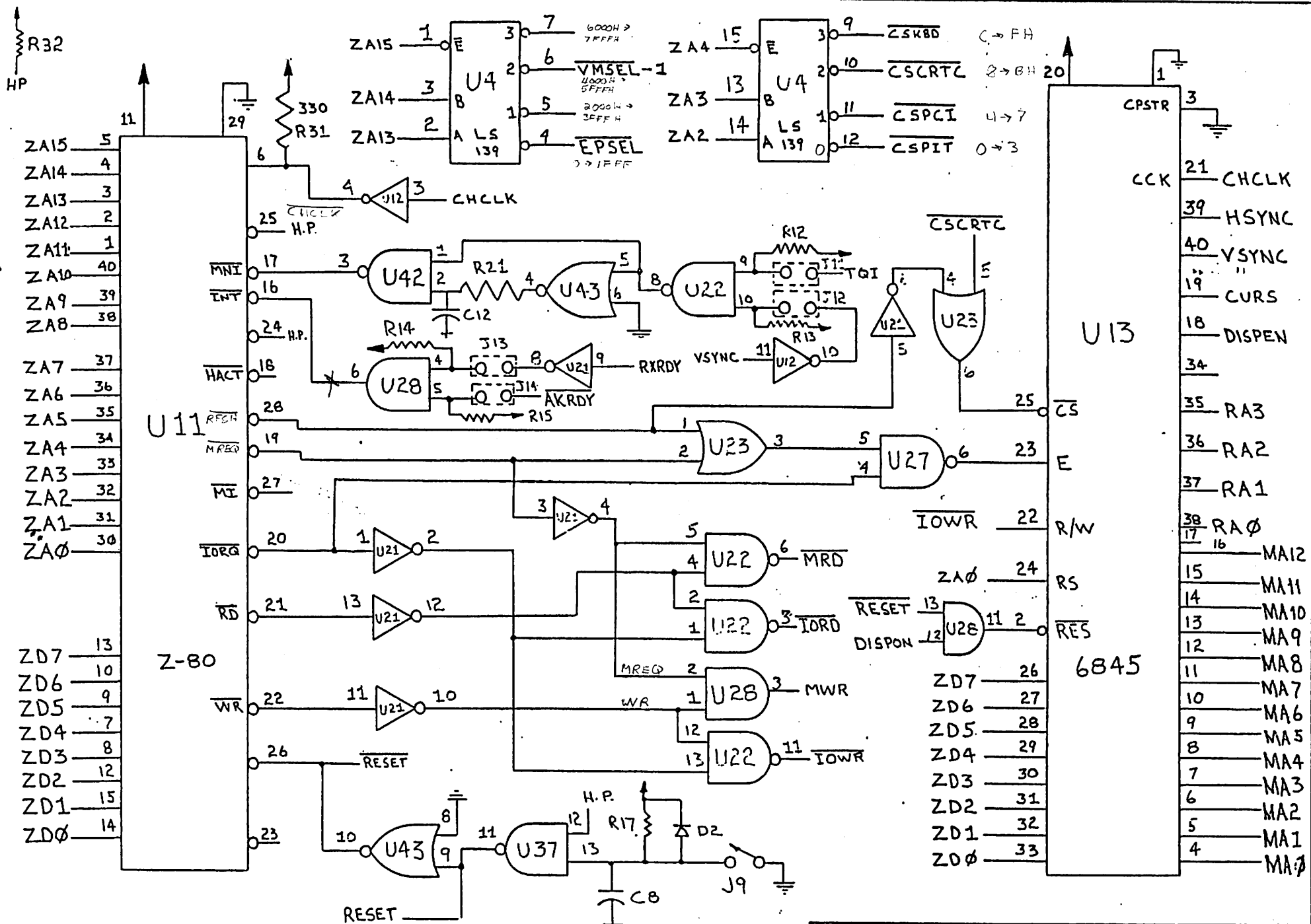
- 0 0 - Non-blink
- 0 1 - No cursor display
- 1 0 - Blink, $1/16$ field rate
- 1 1 - Blink, $1/32$ field rate

CONTROL CODE JUMP TABLE

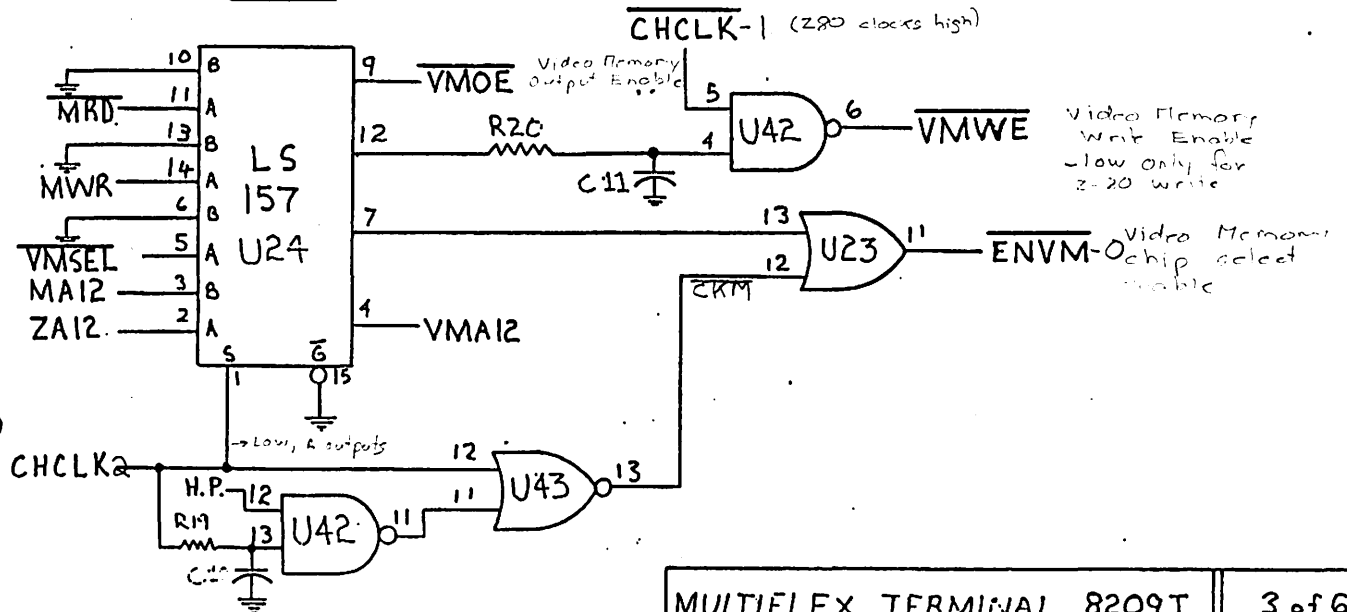
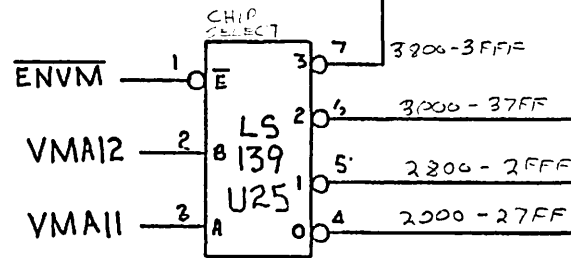
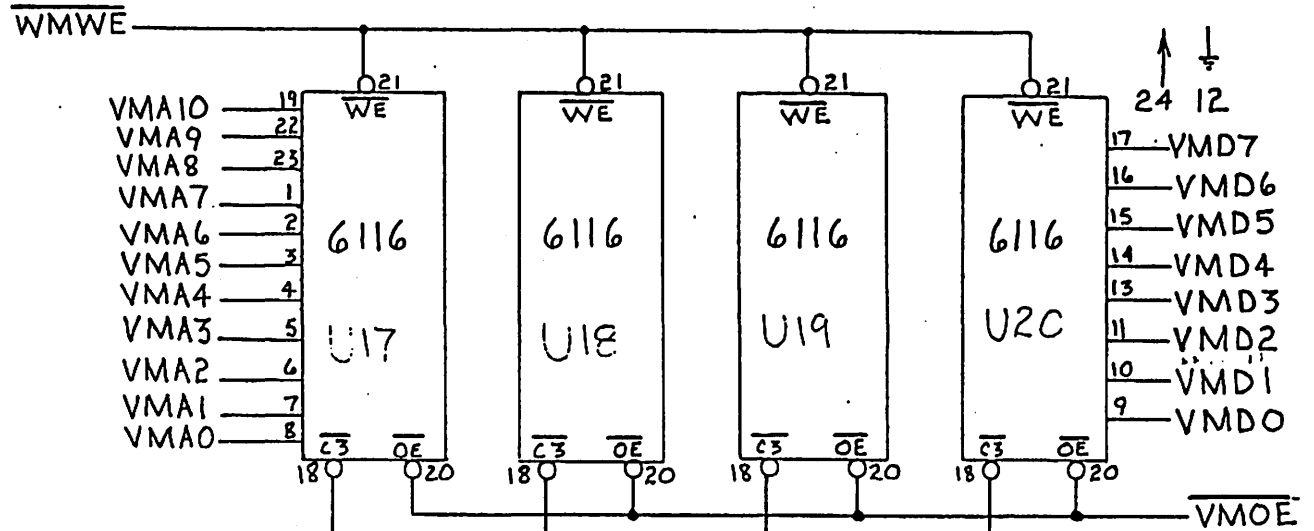
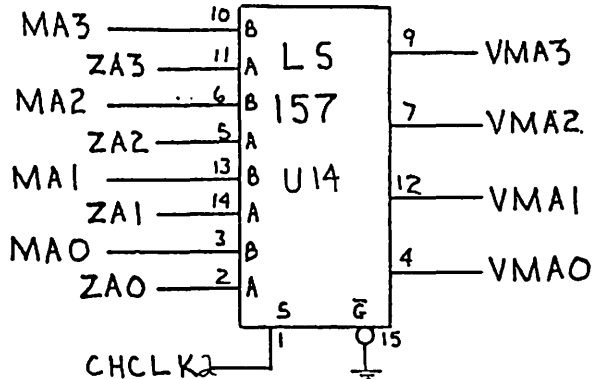
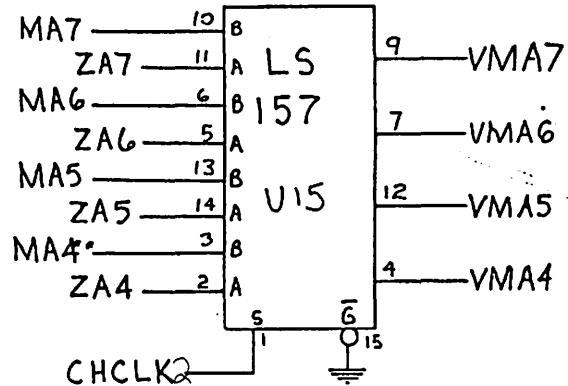
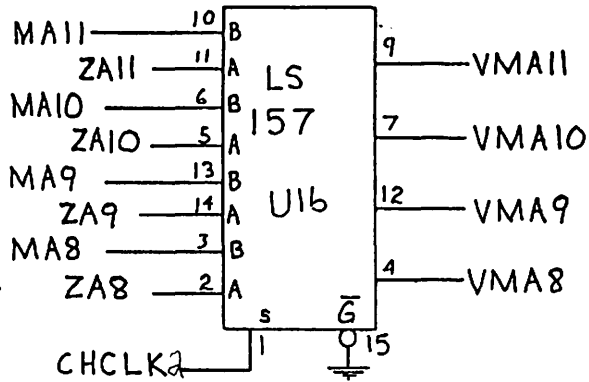
ADDRESS	DATA (JUMP ADDR)	LOCAL MODE:			
04C9	0549	CLJT: DW CNU	; ^@ - NUL - SETUP		
ED. TERMINAL	V1.7	MACRO-80 3.34	31-OCT-79	PAGE	1-14
04CB	0549	CNU	; ^A - SOH		
04CD	0549	CNU	; ^B - STX		
04CF	0617	CNXPB	; ^C - ETX - NEXT PG		
04D1	05F0	CSPA	; ^D - EOT - STOP ATR'S		
04D3	066A	CSCD	; ^E - ENQ - SCROLL DN		
04D5	05EA	CSTA	; ^F - ACK - STRT ATR'S		
04D7	05F5	CBEL	; ^G - BEL		
04D9	054A	CBS	; ^H - BS		
04DB	054E	CTAB	; ^I - HT - TAB		
04DD	0585	CLF	; ^J - LF		
04DF	056A	CUPLN	; ^K - VT - UP LINE		
04E1	0590	CFWSP	; ^L - FF - FWD SPC		
04E3	05A6	CNL	; ^M - CR		
04E5	0549	CNU	; ^N - SO		
04E7	0549	CNU	; ^O - SI		
04E9	05C2	CEOL	; ^P - DLE - END OF LN		
04EB	0549	CNU	; ^Q - DC1		
04ED	0641	CPRPG	; ^R - DC2 - PREV PG		
04EF	0549	CNU	; ^S - DC3 - XMIT STAT		
04F1	067A	TXBU	; ^T - DC4 - XMIT BUF		
04F3	0549	CNU	; ^U - NAK		
04F5	0549	CNU	; ^V - SYN		
04F7	0549	CNU	; ^W - ETB		
04F9	0672	CSCU	; ^X - CAN - SCROLL UP		
04FB	0549	CNU	; ^Y - EM		
04FD	0577	CCLSC	; ^Z - SUB - CLR SCRN		
04FF	05E4	CESC	; ^[- ESC - LOAD CURS		
0501	0549	CNU	; ^\ - FS		
0503	0549	CNU	; ^] - GS - INS LN		
0505	057B	CHOM	; ^^ - RS - HOME CURS		
0507	0549	CNU	; ^_ - VS - DEL LN		
REMOTE MODE:					
0509	0549	CRJT: DW CNU	; ^@ - NUL - SETUP		
050B	0549	DW CNU	; ^A - SOH		
050D	0549	DW CNU	; ^B - STX		
050F	0549	DW CNU	; ^C - ETX		
0511	05F0	DW CSPA	; ^D - EOT - STOP ATR'S		
0513	0549	DW CNU	; ^E - ENQ		
0515	05EA	DW CSTA	; ^F - ACK - STRT ATR'S		
0517	05F5	DW CBEL	; ^G - BEL		
0519	054A	DW CBS	; ^H - BS		
051B	054E	DW CTAB	; ^I - HT - TAB		
051D	0585	DW CLF	; ^J - LF		
051F	056A	DW CUPLN	; ^K - VT - UP LINE		
0521	0590	DW CFWSP	; ^L - FF - FWD SPC		
0523	05A6	DW CNL	; ^M - CR		
VIDEO TERMINAL	V1.7	MACRO-80 3.34	31-OCT-79	PAGE	1-15

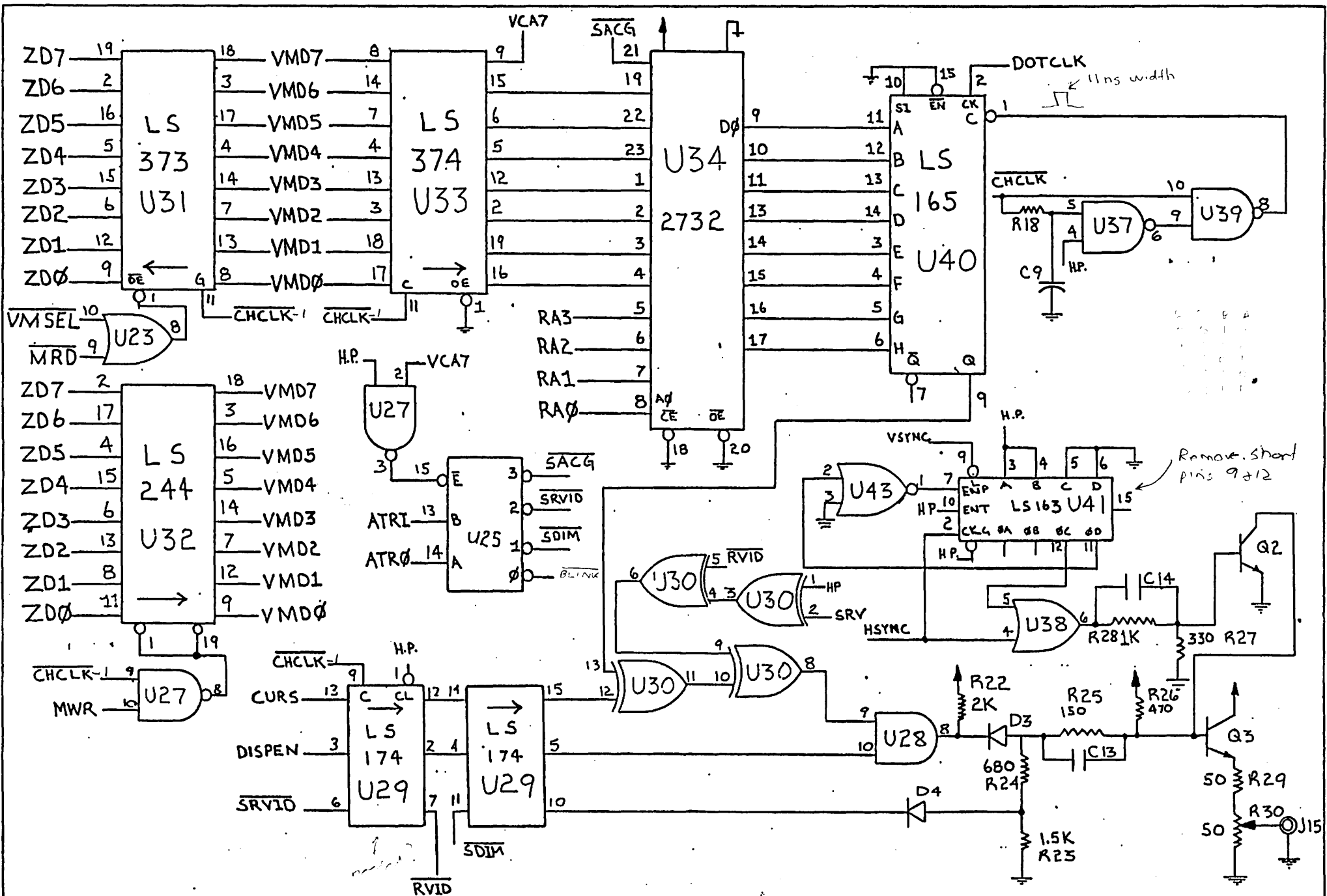
0525	0549	DW	CNU	;	^N - SO
0527	0549	DW	CNU	;	^O - SI
0529	0502	DW	CEOL	;	^P - DLE - END OF LN
2B	0549	DW	CNU	;	^Q - DC1
052D	0549	DW	CNU	;	^R - DC2
052F	0549	DW	CNU	;	^S - DC3
0531	0549	DW	CNU	;	^T - DC4
0533	0549	DW	CNU	;	^U - NAK
0535	0549	DW	CNU	;	^V - SYN
0537	0549	DW	CNU	;	^W - ETB
0539	0549	DW	CNU	;	^X - CAN
053B	0549	DW	CNU	;	^Y - EM
053D	0577	DW	CCLSC	;	^Z - SUB - CLR SCRN
053F	05E4	DW	CESC	;	^[- ESC - LOAD CURS
0541	0549	DW	CNU	;	^_ - FS
0543	0549	DW	CNU	;	^] - GS
0545	057B	DW	CHOM	;	^^ - RS - HOME CURS
0547	0549	DW	CNU	;	^_ - VS - BREAK

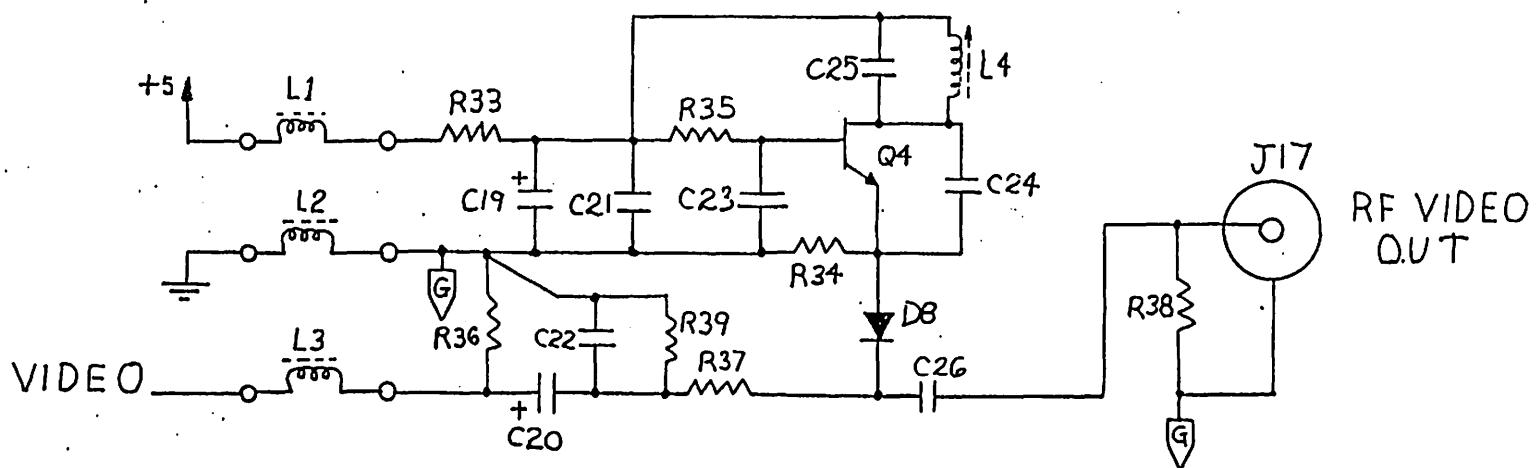
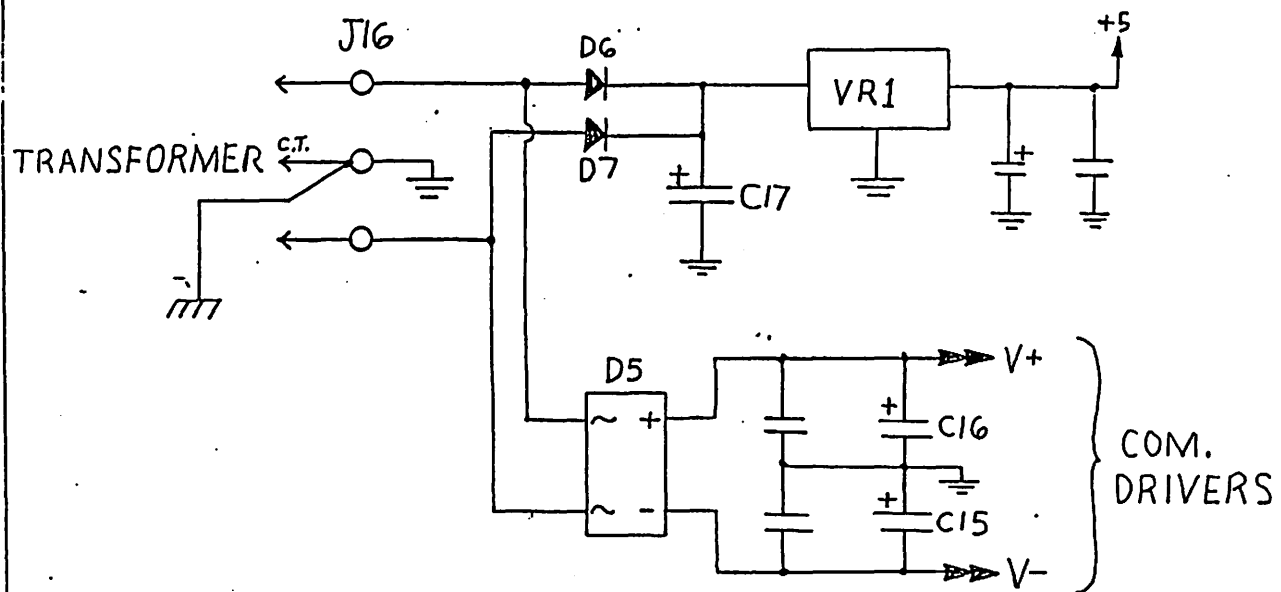
93 C9 CNU: RET



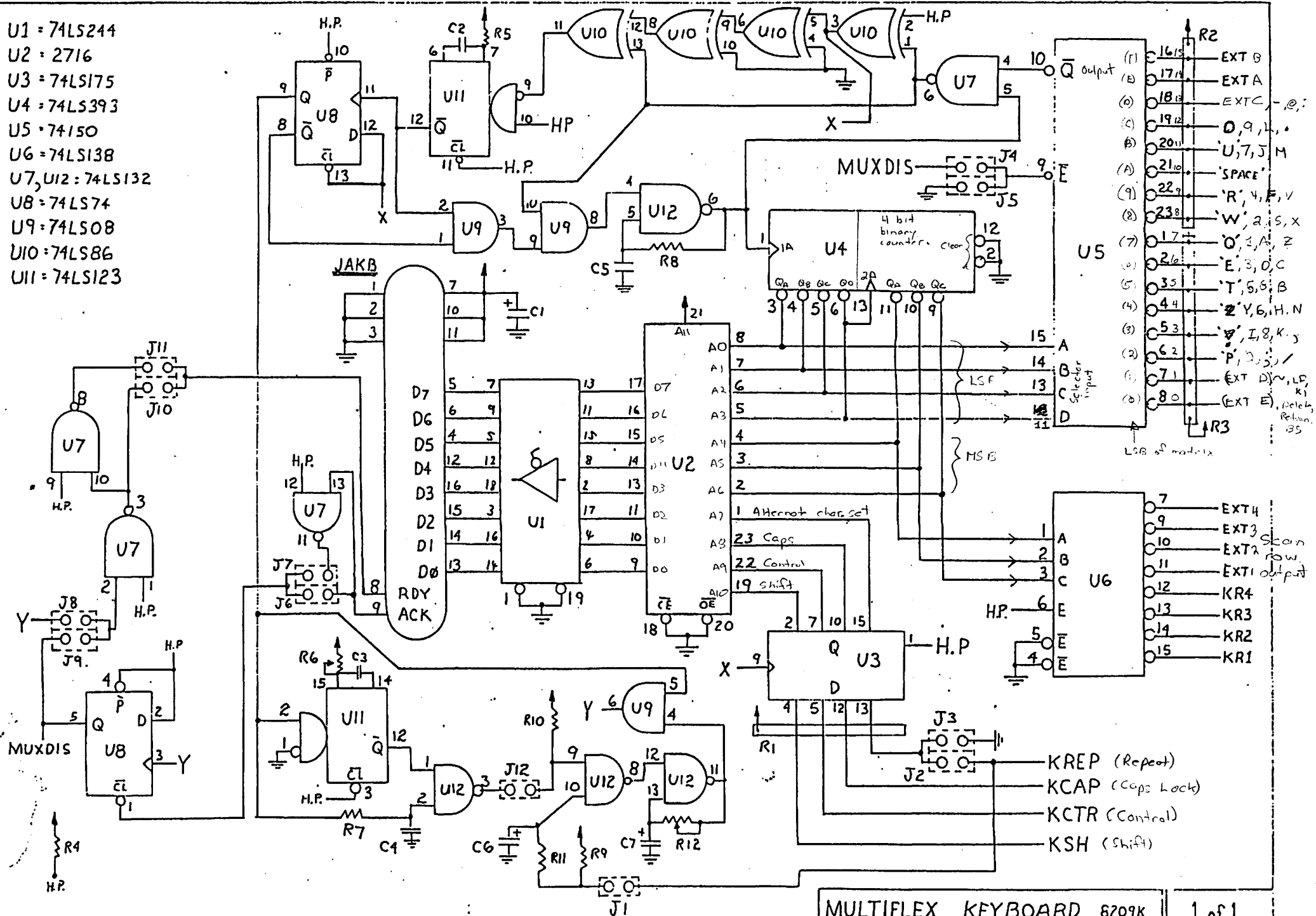








U1 = 74LS244
 U2 = 2716
 U3 = 74LS175
 U4 = 74LS393
 U5 = 74150
 U6 = 74LS138
 U7, U12 = 74LS132
 U8 = 74LS74
 U9 = 74LS08
 U10 = 74LS86
 U11 = 74LS123



PARTS LIST

TERMINAL SECTION

IC's

[]	1	74S00	U39
[]	2	74LS00	U22, U27
[]	1	74LS02	U43
[]	2	74LS04	U12, U21
[]	1	74LS08	U28
[]	3	74LS32	U23, U26, U38
[]	1	74LS86	U30
[]	1	74S124	U35
[]	2	74LS132	U37, U42
[]	2	74LS139	U4, U25
[]	4	74LS157	U14, U15, U16, U24
[]	1	74S163 or 74LS163	U36
[]	1	74LS163	U41
[]	1	74LS165	U40
[]	2	74LS174	U6, U29
[]	1	74LS240	U7
[]	2	74LS244	U8, U32
[]	1	74LS373	U31
[]	1	74LS374	U33
[]	2	1489	U1, U2
[]	1	1488	U3
[]	1	Z80-CPU	U11
[]	1	6845/6545	U13
[]	1	8251	U9
[]	2	2732	U5 ("MON"), U34 ("CHAR. GEN")
[]	1	8253	U10
[]	2	6116/2016	U17, U20

Resistors:

[]	1	20R	R2
[]	1	47R	R3
[]	1	50R	R29
[]	1	150R	R25
[]	6	330R	R18-21, R27, R31
[]	1	470R	R26
[]	1	680R	R24
[]	3	1K	R8, R9, R28
[]	1	1.5K	R23
[]	9	2K	R4, R5, R12-16, R22, R32
[]	3	5.6K	R40-42
[]	3	10K	R6, R7, R17
[]	1	50R vertical trimpot	R1
[]	1	50R " "	R30
[]	1	2.2K 8 pin SIP	R11

Capacitors:

[]	1	10 pF	C4
[]	1	33 pF	C9
[]	1	68 pF	C11
[]	3	100 pF	C5, C10, C12
[]	1	470 pF	C18
[]	1	1 nF	C3
[]	31	.1 μ F	C1, C2, and 29 bypass caps.
[]	3	10 μ F tant.	C6, C7, C27
[]	1	22 μ F "	C8

Misc. Components:

[]	1	MPSA13	Q1
[]	2	2N4401, PN2222	Q2, Q3
[]	4	1N914	D1-4
[]	1	DB25S	JPC1
[]	1	2 MHz XTAL	Y1
[]	1	10 MHz XTAL	Y2

Sockets:

[]	2	40 pin
[]	2	28 pin
[]	6	24 pin
[]	5	20 pin
[]	13	16 pin
[]	16	14 pin

KEYBOARD ENCODER

IC's:

[]	1	74LS08	U9
[]	1	74LS74	U8
[]	1	74LS86	U10
[]	1	74LS123	U11
[]	2	74LS132	U7, U12
[]	1	74LS138	U6
[]	1	74150	U5
[]	1	74LS175	U3
[]	1	74LS244	U1
[]	1	74LS393	U4
[]	1	2716	U2 ("KYBD ENC.")

Resistors:

[]	2	330R	R7, R8
[]	2	1K	R10, R11
[]	2	2K	R9, R4
[]	1	82K	R5
[]	1	1K Horiz. trimpot	R12

[]	1	220K Vert.	"	R6
[]	2	2.2K 10 pin SIP		R2, R3
[]	1	2.2K 8 pin SIP		R1

Capacitors:

[]	1	1 nF		C4
[]	13	.1 μ F		C2 (2), C5, and 10 bypass
[]	1	10 μ F tant.		C1
[]	1	15 μ F	"	C6
[]	1	47 μ F elec.		C3
[]	1	220 μ F	"	C7

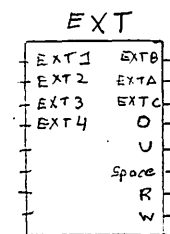
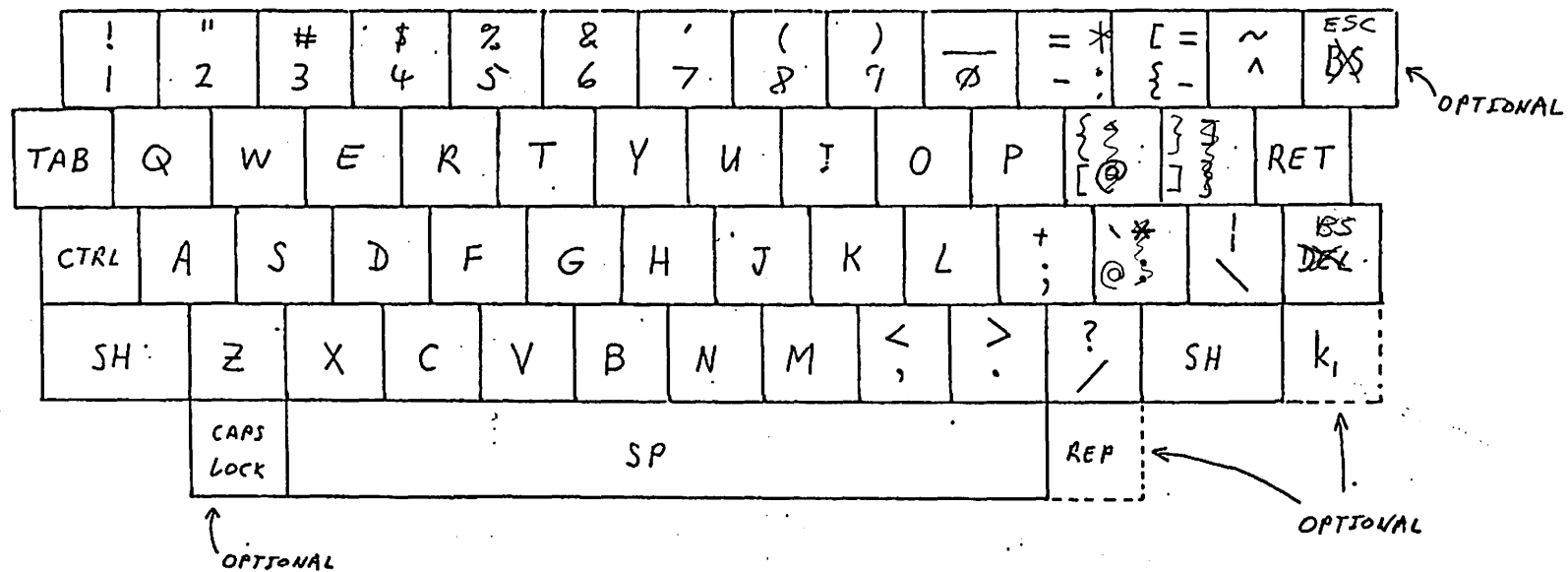
Sockets:

[]	2	24 pin
[]	1	20 pin
[]	4	16 pin
[]	6	14 pin

OPTIONS:

[]	1	Speaker
[]	4	Keyswitch c/w caps
[]	2	6116/2016
[]	1	DB25S
[]	1	Power supply parts kit
[]	1	RF modulator parts kit
[]	1	Reset Pushbutton

KEYBOARD



KEY BOARD MATRIX

		EXTB	EXTA	EXTC	O	U	Space	R	W
EXT4	7								
MJD	EXT3	6							
	EXT2	5							
	EXT1	4							
KR4	3	Z	C	B	N	<	?	k ₁	
KR3	2	A	D	G	H	K	+		BS DEL
KR2	1	Q	E	T	Y	I	P	CR	
KR1	0	!	#	%	&	(	_	E	~
		1	3	5	6	8	0	{	^
		7	6	5	4	3	2	1	0
		F	E	D	C	B	A	9	8

LSD

2716

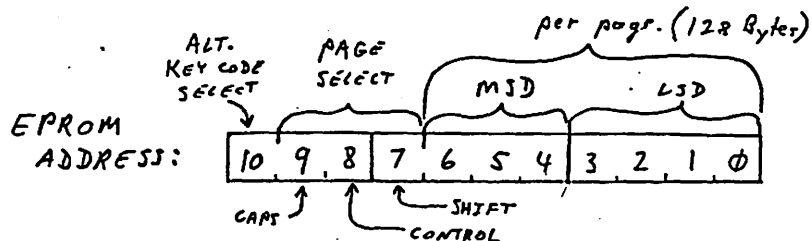
EPROM PAGES

0
20
100
180
200
220
240
260
280
300
320
340
360
380
400
420
440
460
480
500
520
540
560
580
600
620
640
660
680
700
720
740
760
780

ALT
KEY
CODES

- 3 - caps, control, shift } Fin key changes
- 3 - caps, control
- 2 - caps, shift
- 4 - caps
- 3 - control, shift } Fin key changes
- 3 - Control
- 2 - Shift

Page



-ALL UNENCODED KEYS (BLANK ENTRES) ARE FF.H.

Function keys

F1	Shift Ctrl	1
F2		2
F3		3
F4		4
F5		5
Blue		6
Red		7
Gray		8

Output
from
Terminal
Esc S

Output From Keyboard
F0h
F1h
F2h
F3h
F4h
F5h
F6h
F7h

→ Shift-control-ESC or Shift-caps-control ESC enters
Setup mode if in remote mode (Keyboard sends E0h)

STANDARD (LOWER CASE + NUMBERS)

1

80	5E	Λ
81	7B	Σ
82	30	∅
83	38	8
84	36	6
85	35	5
86	33	3
87	31	1
88	32	2
89	34	4
8A	08	BS
8B	37	7
8C	39	9
8D	2D	-
8E		
8F		
90	0D	RET
91	7D	3
92	70	P
93	69	i
94	79	y
95	74	t
96	65	e
97	71	q
98	77	w
99	72	r
9A	09	TAB
9B	75	u
9C	6F	o
9D	40	@
9E		
9F		

A0	7F	DEL
A1	5C	\
A2	3B	;
A3	6B	k
A4	68	h
A5	67	g
A6	64	α
A7	61	α
A8	73	α
A9	66	F
AA		
AB	6A	j
AC	6C	l
AD	3A	:
AE		
AF		
B0		
B1	FF	k ₁
B2	2F	/
B3	26	,
B4	6E	n
B5	62	h
B6	63	c
B7	7A	z
B8	78	x
B9	76	v
BA	20	SP
BB	6D	m
BC	2E	.
BD		
BE		
BF		

C0		
C1		
C2		
C3		
C4		
C5		
C6		
C7		
C8		
C9		
CA		
CB		
CC		
CD		
CE		
CF		
D0		
D1		
D2		
D3		
D4		
D5		
D6		
D7		
D8		
D9		
DA		
DB		
DC		
DD		
DE		
DF		

E0		
E1		
E2		
E3		
E4		
E5		
E6		
E7		
EP		
E9		
EA		
EB		
EC		
ED		
EE		
EF		
F0		
F1		
F2		
F3		
F4		
F5		
F6		
F7		
F8		
F9		
FA		
FB		
FC		
FD		
FE		
FF		

SHIFT (UPPER CASE & SYMBOLS)

00	7E	~
01	5B	£
02	5F	—
03	28	(
04	26	&
05	25	%
06	23	#
07	21	!
08	22	"
09	24	\$
0A	08	BS
0B	27	'
0C	29	)
0D	3D	=
0E		
0F		
10	0D	CR
11	53	]
12	50	P
13	49	I
14	59	Y
15	54	T
16	45	E
17	51	Q
18	57	W
19	52	R
1A	09	TAB
1B	55	U
1C	4F	O
1D	60	.
1E		
1F		

20	7F	DEL
21	7C	
22	28	+
23	48	K
24	48	H
25	47	G
26	44	D
27	41	A
28	53	S
29	46	F
2A		
2B	4A	J
2C	4C	L
2D	2A	*
2E		
2F		
30		
31	FF	k
32	3F	?
33	3C	<
34	4E	N
35	42	B
36	43	C
37	5A	Z
38	58	X
39	56	V
3A	20	SP
3B	4D	M
3C	3E	>
3D		
3E		
3F		

40		
41		
42		
43		
44		
45		
46		
47		
48		
49		
4A		
4B		
4C		
4D		
4E		
4F		
50		
51		
52		
53		
54		
55		
56		
57		
58		
59		
5A		
5B		
5C		
5D		
5E		
5F		

60		
61		
62		
63		
64		
65		
66		
67		
68		
69		
6A		
6B		
6C		
6D		
6E		
6F		
70		
71		
72		
73		
74		
75		
76		
77		
78		
79		
7A		
7B		
7C		
7D		
7E		
7F		

CONTROL (CONTROL CODES & NUMBERS)

3

80	1E	RS
81	1B	ESC
82	1F	US
83	38	8
84	36	6
85	35	5
86	33	3
87	31	1
88	32	2
89	34	4
8A	08	BS
8B	37	7
8C	39	9
8D	2D	-
8E		
8F		
90	0D	CR
91	1D	GS
92	10	DLE
93	09	HT
94	19	EM
95	14	DC4
96	05	ENR
97	11	DC1
98	17	ETB
99	12	DC2
9A	09	TAB
9B	15	NAK
9C	0F	SI
9D	00	NUL
9E		
9F		

A0	7F	DEL
A1	1C	FJ
A2	3B	;
A3	0B	VT
A4	08	BS
A5	07	BEL
A6	04	EOT
A7	01	SOH
A8	13	DC3
A9	06	ACK
AA		
AB	0A	LF
AC	0C	FF
AD	3A	:
AE		
AF		
B0		
B1	FF	K1
B2	2F	/
B3	2C	,
B4	0E	SO
B5	02	STX
B6	03	ETX
B7	1A	SUB
B8	18	CAN
B9	16	SYN
BA	20	SP
BB	0D	CR
BC		
BD		
BE		
BF		

C0		
C1		
C2		
C3		
C4		
C5		
C6		
C7		
C8		
C9		
CA		
CB		
CC		
CD		
CE		
CF		
D0		
D1		
D2		
D3		
D4		
D5		
D6		
D7		
D8		
D9		
DA		
DB		
DC		
DD		
DE		
DF		

E0		
E1		
E2		
E3		
E4		
E5		
E6		
E7		
E8		
E9		
EA		
EB		
EC		
ED		
EE		
EF		
F0		
F1		
F2		
F3		
F4		
F5		
F6		
F7		
F8		
F9		
FA		
FB		
FC		
FD		
FE		
FF		

CAPS (UPPER CASE & NUMBERS)

80	5E	1
81	7B	{
82	30	0
83	38	8
84	36	6
85	35	5
86	33	3
87	31	1
88	32	2
89	34	4
8A	08	BS
8B	37	7
8C	39	9
8D	23	-
8E		
8F		
90	0D	CR
91	7D	}
92	50	P
93	49	I
94	59	Y
95	54	T
96	45	E
97	51	Q
98	57	W
99	52	R
9A	09	TAB
9B	55	U
9C	4F	O
9D	40	@
9E		
9F		

A0	7F	DEL
A1	5C	\
A2	3B	3
A3	4B	K
A4	48	H
A5	47	G
A6	44	D
A7	41	A
A8	53	S
A9	46	F
AA		.
AB	4A	J
AC	4C	L
AD	3A	:
AE		
AF		
B0		
B1	FF	K1
B2	2F	/
B3	2C	>
B4	4E	N
B5	42	B
B6	43	C
B7	5A	Z
B8	58	X
B9	56	V
BA	20	SP
BB	4D	M
BC	2E	.
BD		
BE		
BF		

C0		
C1		
C2		
C3		
C4		
C5		
C6		
C7		
C8		
C9		
CA		
CB		
CC		
CD		
CE		
CF		
D0		
D1		
D2		
D3		
D4		
D5		
D6		
D7		
D8		
D9		
DA		
DB		
DC		
DD		
DE		
DF		

E0		
E1		
E2		
E3		
E4		
E5		
E6		
E7		
EF		
E9		
EA		
EB		
EC		
ED		
EE		
EF		
F0		
F1		
F2		
F3		
F4		
F5		
F6		
F7		
F8		
F9		
FA		
FB		
FC		
FD		
FE		
FF		

TERM
CG

00	00	SP
01	00	
02	00	
03	00	
04	00	
05	00	
06	00	
07	00	
08	00	
09	00	
0A	00	
0B		
0C		
0D		
0E		
0F		
10	00	!
11	10	
12	10	
13	10	
14	10	
15	10	
16	00	
17	10	
18	00	
19	00	
1A	00	
1B		
1C		
1D		
1E		
1F		

20	00	
21	28	
22	28	
23	28	
24	00	
25	00	
26	00	
27	00	
28	00	
29	00	
2A	00	
2B		
2C		
2D		
2E		
2F		
30	00	#
31	28	
32	28	
33	7C	
34	28	
35	7C	
36	28	
37	28	
38	00	
39	00	
3A	00	
3B		
3C		
3D		
3E		
3F		

40	00	\$
41	10	
42	3C	
43	50	
44	38	
45	14	
46	78	
47	10	
48	00	
49	00	
4A	00	
4B		
4C		
4D		
4E		
4F		
50	00	%
51	64	
52	64	
53	08	
54	10	
55	20	
56	4C	
57	4C	
58	00	
59	00	
5A	00	
5B		
5C		
5D		
5E		
5F		

60	00	&
61	20	
62	50	
63	50	
64	20	
65	54	
66	48	
67	34	
68	00	
69	00	
6A	00	
6B		
6C		
6D		
6E		
6F		
70	00	!
71	10	
72	10	
73	10	
74	00	
75	00	
76	00	
77	00	
78	00	
79	00	
7A	00	
7B		
7C		
7D		
7E		
7F		

OA

6

TERM
CG

80	00	(
81	08	
82	10	
83	20	
84	20	
85	20	
86	10	
87	08	
88	00	
89	00	
8A	00	
8B		
8C		
8D		
8E		
8F		
90	00	)
91	20	
92	10	
93	08	
94	08	
95	08	
96	10	
97	20	
98	00	
99	00	
9A	00	
9B		
9C		
9D		
9E		
9F		

A0	00	*
A1	10	
A2	54	
A3	38	
A4	10	
A5	38	
A6	54	
A7	10	
A8	00	
A9	00	
AA	00	
AB		
AC		
AD		
AE		
AF		
B0	00	+
B1	00	
B2	10	
B3	10	
B4	7C	
B5	10	
B6	10	
B7	00	
B8	00	
B9	00	
BA	00	
BB		
BC		
BD		
BE		
BF		

C0	00	,
C1	00	
C2	00	
C3	00	
C4	00	
C5	00	
C6	10	
C7	10	
C8	20	
C9	00	
CA	00	
CB		
CC		
CD		
CE		
CF		
D0	00	-
D1	00	
D2	00	
D3	00	
D4	7C	
D5	00	
D6	00	
D7	00	
D8	00	
D9	00	
DA	00	
DB		
DC		
DD		
DE		
DF		

E0	00	
E1	00	
E2	00	
E3	00	
E4	00	
E5	00	
E6	00	
E7	10	
E8	00	
E9	00	
EA	00	
EB		
EC		
ED		
EE		
EF		
F0	00	/
F1	00	
F2	04	
F3	08	
F4	10	
F5	20	
F6	40	
F7	00	
F8	00	
F9	00	
FA	00	
FB		
FC		
FD		
FE		
FF		

OB

00	00	Ø
01	38	
02	44	
03	4C	
04	54	
05	64	
06	44	
07	38	
08	00	
09	00	
0A	00	
0B		
0C		
0D		
0E		
0F		
10	00	1
11	10	
12	30	
13	10	
14	10	
15	10	
16	10	
17	38	
18	00	
19	00	
1A	00	
1B		
1C		
1D		
1E		
1F		

20	00	2
21	38	
22	44	
23	04	
24	18	
25	20	
26	40	
27	7C	
28	00	
29	00	
2A	00	
2B		
2C		
2D		
2E		
2F		
30	00	3
31	7C	
32	04	
33	08	
34	18	
35	04	
36	44	
37	38	
38	00	
39	00	
3A	00	
3B		
3C		
3D		
3E		
3F		

40	00	4
41	08	
42	18	
43	28	
44	48	
45	7C	
46	08	
47	08	
48	00	
49	00	
4A	00	
4B		
4C		
4D		
4E		
4F		
50	00	5
51	7C	
52	40	
53	78	
54	04	
55	04	
56	44	
57	38	
58	00	
59	00	
5A	00	
5B		
5C		
5D		
5E		
5F		

60	00	6
61	1C	
62	20	
63	40	
64	78	
65	44	
66	44	
67	38	
68	00	
69	00	
6A	00	
6B		
6C		
6D		
6E		
6F		
70	00	7
71	7C	
72	04	
73	08	
74	10	
75	20	
76	20	
77	20	
78	00	
79	00	
7A	00	
7B		
7C		
7D		
7E		
7F		

TERM
CG

OB

80	00	8
81	38	
82	44	
83	44	
84	38	
85	44	
86	44	
87	38	
88	00	
89	00	
8A	00	
8B		
8C		
8D		
8E		
8F		
90	00	9
91	38	
92	44	
93	44	
94	3C	
95	04	
96	08	
97	70	
98	00	
99	00	
9A	00	
9B		
9C		
9D		
9E		
9F		

A0	00	:
A1	00	
A2	00	
A3	10	
A4	00	
A5	00	
A6	10	
A7	00	
A8	00	
A9	00	
AA	00	
AB		
AC		
AD		
AE		
AF		
B0	00	;
B1	00	
B2	00	
B3	00	
B4	10	
B5	00	
B6	10	
B7	10	
B8	20	
B9	00	
BA	00	
BB		
BC		
BD		
BE		
BF		

C0	00	<
C1	08	
C2	10	
C3	20	
C4	40	
C5	20	
C6	10	
C7	08	
C8	00	
C9	00	
CA	00	
CB		
CC		
CD		
CE		
CF		
D0	00	=
D1	00	
D2	00	
D3	7C	
D4	00	
D5	7C	
D6	00	
D7	00	
D8	00	
D9	00	
DA	00	
DB		
DC		
DD		
DE		
DF		

E0	00	>
E1	20	
E2	10	
E3	08	
E4	04	
E5	08	
E6	10	
E7	20	
E8	00	
E9	00	
EA	00	
EB		
EC		
ED		
EE		
EF		
F0	00	?
F1	38	
F2	44	
F3	08	
F4	10	
F5	10	
F6	00	
F7	10	
F8	00	
F9	00	
FA	00	
FB		
FC		
FD		
FE		
FF		

TERM

CG

8

TERM

CG

00	00	@
01	38	
02	44	
03	5c	
04	54	
05	5c	
06	40	
07	3c	
08	00	
09	00	
0A	00	
0B		
0C		
0D		
0E		
0F		
10	00	A
11	38	
12	44	
13	44	
14	7c	
15	44	
16	44	
17	44	
18	00	
19	00	
1A	00	
1B		
1C		
1D		
1E		
1F		

20	00	B
21	78	
22	24	
23	24	
24	38	
25	24	
26	24	
27	78	
28	00	
29	00	
2A	00	
2B		
2C		
2D		
2E		
2F		
30	00	C
31	38	
32	44	
33	40	
34	40	
35	40	
36	44	
37	38	
38	00	
39	00	
3A	00	
3B		
3C		
3D		
3E		
3F		

40	00	D
41	78	
42	24	
43	24	
44	24	
45	24	
46	24	
47	78	
48	00	
49	00	
4A	00	
4B		
4C		
4D		
4E		
4F		
50	00	E
51	7c	
52	40	
53	40	
54	78	
55	40	
56	40	
57	7c	
58	00	
59	00	
5A	00	
5B		
5C		
5D		
5E		
5F		

60	00	F
61	7c	
62	40	
63	40	
64	78	
65	40	
66	40	
67	40	
68	00	
69	00	
6A	00	
6B		
6C		
6D		
6E		
6F		
70	00	G
71	3c	
72	40	
73	40	
74	5c	
75	44	
76	44	
77	38	
78	00	
79	00	
7A	00	
7B		
7C		
7D		
7E		
7F		

00

10

TERM

CG

80	00	H
81	44	
82	44	
83	44	
84	7C	
85	44	
86	44	
87	44	
88	00	
89	00	
8A	00	
8B		
8C		
8D		
8E		
8F		
90	00	I
91	38	
92	10	
93	10	
94	10	
95	10	
96	10	
97	38	
98	00	
99	00	
9A	00	
9B		
9C		
9D		
9E		
9F		

A0	00	J
A1	04	
A2	04	
A3	04	
A4	04	
A5	04	
A6	44	
A7	38	
A8	00	
A9	00	
AA	00	
AB		
AC		
AD		
AE		
AF		
B0	00	K
B1	44	
B2	48	
B3	50	
B4	60	
B5	50	
B6	48	
B7	44	
B8	00	
B9	00	
BA	00	
BB		
BC		
BD		
BE		
BF		

C0	00	L
C1	40	
C2	40	
C3	40	
C4	40	
C5	40	
C6	40	
C7	7C	
C8	00	
C9	00	
CA	00	
CB		
CC		
CD		
CE		
CF		
D0	00	M
D1	44	
D2	6C	
D3	54	
D4	54	
D5	44	
D6	44	
D7	44	
D8	00	
D9	00	
DA	00	
DB		
DC		
DD		
DE		
DF		

E0	00	N
E1	44	
E2	64	
E3	54	
E4	4C	
E5	44	
E6	44	
E7	44	
E8	00	
E9	00	
EA	00	
EB		
EC		
ED		
EE		
EF		
F0	00	O
F1	32	
F2	44	
F3	44	
F4	44	
F5	44	
F6	44	
F7	38	
F8	00	
F9	00	
FA	00	
FB		
FC		
FD		
FE		
FF		

0D

00	00	P
01	78	
02	44	
03	44	
04	78	
05	40	
06	40	
07	40	
08	00	
09	00	
0A	00	
0B		
0C		
0D		
0E		
0F		
10	00	Q
11	38	
12	44	
13	44	
14	44	
15	54	
16	48	
17	34	
18	00	
19	00	
1A	00	
1B		
1C		
1D		
1E		
1F		

20	00	R
21	78	
22	44	
23	44	
24	78	
25	50	
26	48	
27	44	
28	00	
29	00	
2A	00	
2B		
2C		
2D		
2E		
2F		
30	00	S
31	38	
32	44	
33	40	
34	38	
35	04	
36	44	
37	38	
38	00	
39	00	
3A	00	
3B		
3C		
3D		
3E		
3F		

40	00	T
41	7C	
42	10	
43	10	
44	10	
45	10	
46	10	
47	10	
48	00	
49	00	
4A	00	
4B		
4C		
4D		
4E		
4F		
50	00	U
51	44	
52	44	
53	44	
54	44	
55	44	
56	44	
57	38	
58	00	
59	00	
5A	00	
5B		
5C		
5D		
5E		
5F		

60	00	V
61	44	
62	44	
63	44	
64	44	
65	44	
66	28	
67	10	
68	00	
69	00	
6A	00	
6B		
6C		
6D		
6E		
6F		
70	00	W
71	44	
72	44	
73	44	
74	54	
75	54	
76	6C	
77	54	
78	00	
79	00	
7A	00	
7B		
7C		
7D		
7E		
7F		

TERM

CG

TERM

CG

80	00	X
81	44	
82	44	
83	28	
84	10	
85	28	
86	44	
87	44	
88	00	
89	00	
8A	00	
8B		
8C		
8D		
8E		
8F		
90	00	V
91	44	
92	44	
93	28	
94	10	
95	10	
96	10	
97	10	
98	00	
99	00	
9A	00	
9B		
9C		
9D		
9E		
9F		

A0	00	Z
A1	7C	
A2	04	
A3	08	
A4	10	
A5	20	
A6	40	
A7	7C	
A8	00	
A9	00	
AA	00	
AB		
AC		
AD		
AE		
AF		
B0	00	[
B1	3C	
B2	20	
B3	20	
B4	20	
B5	20	
B6	20	
B7	3C	
B8	00	
B9	00	
BA	00	
BB		
BC		
BD		
BE		
BF		

C0	00	\
C1	00	
C2	40	
C3	20	
C4	10	
C5	08	
C6	04	
C7	00	
C8	00	
C9	00	
CA	00	
CB		
CC		
CD		
CE		
CF		
D0	00	]
D1	78	
D2	08	
D3	08	
D4	08	
D5	08	
D6	08	
D7	78	
D8	00	
D9	00	
DA	00	
DB		
DC		
DD		
DE		
DF		

E0	00	^
E1	00	
E2	00	
E3	10	
E4	28	
E5	44	
E6	00	
E7	00	
E8	00	
E9	00	
EA	00	
EB		
EC		
ED		
EE		
EF		
F0	00	—
F1	00	
F2	00	
F3	00	
F4	00	
F5	00	
F6	00	
F7	00	
F8	7C	
F9	00	
FA	00	
FB		
FC		
FD		
FE		
FF		

OE

00	00	
01	20	
02	10	
03	08	
04	00	
05	00	
06	00	
07	00	
08	00	
09	00	
0A	00	
0B		
0C		
0D		
0E		
0F		
10	00	α
11	00	
12	00	
13	38	
14	04	
15	3C	
16	44	
17	3C	
18	00	
19	00	
1A	00	
1B		
1C		
1D		
1E		
1F		

20	00	b
21	40	
22	40	
23	58	
24	64	
25	44	
26	64	
27	58	
28	00	
29	00	
2A	00	
2B		
2C		
2D		
2E		
2F		
30	00	α
31	00	
32	00	
33	3C	
34	40	
35	40	
36	40	
37	3C	
38	00	
39	00	
3A	00	
3B		
3C		
3D		
3E		
3F		

40	00	d
41	04	
42	04	
43	34	
44	4C	
45	44	
46	4C	
47	34	
48	00	
49	00	
4A	00	
4B		
4C		
4D		
4E		
4F		
50	00	e
51	00	
52	00	
53	38	
54	44	
55	78	
56	40	
57	38	
58	00	
59	00	
5A	00	
5B		
5C		
5D		
5E		
5F		

60	00	f
61	18	
62	24	
63	20	
64	70	
65	20	
66	20	
67	20	
68	00	
69	00	
6A	00	
6B		
6C		
6D		
6E		
6F		
70	00	g
71	00	
72	00	
73	34	
74	4C	
75	4C	
76	34	
77	04	
78	38	
79	00	
7A	00	
7B		
7C		
7D		
7E		
7F		

TERM
CG

OE

80	00	h
81	40	
82	40	
83	40	
84	58	
85	64	
86	44	
87	44	
88	00	
89	00	
8A	00	
8B		
8C		
8D		
8E		
8F		
90	00	i
91	00	
92	10	
93	00	
94	30	
95	10	
96	10	
97	38	
98	00	
99	00	
9A	00	
9B		
9C		
9D		
9E		
9F		

A0	00	j
A1	08	
A2	00	
A3	08	
A4	08	
A5	08	
A6	08	
A7	48	
A8	30	
A9	00	
AA	00	
AB		
AC		
AD		
AE		
AF		
B0	00	k
B1	40	
B2	40	
B3	48	
B4	50	
B5	70	
B6	48	
B7	44	
B8	00	
B9	00	
BA	00	
BB		
BC		
BD		
BE		
BF		

C0	00	l
C1	30	
C2	10	
C3	10	
C4	10	
C5	10	
C6	10	
C7	38	
C8	00	
C9	00	
CA	00	
CB		
CC		
CD		
CE		
CF		
D0	00	m
D1	00	
D2	00	
D3	68	
D4	54	
D5	54	
D6	54	
D7	54	
D8	00	
D9	00	
DA	00	
DB		
DC		
DD		
DE		
DF		

E0	00	n
E1	00	
E2	00	
E3	78	
E4	44	
E5	44	
E6	44	
E7	44	
E8	00	
E9	00	
EA	00	
EB		
EC		
ED		
EE		
EF		
F0	00	o
F1	00	
F2	00	
F3	38	
F4	44	
F5	44	
F6	44	
F7	38	
F8	00	
F9	00	
FA	00	
FB		
FC		
FD		
FE		
FF		

TERM

CG

OF

15

TERM
CG

00	00	p
01	00	
02	00	
03	58	
04	64	
05	64	
06	58	
07	40	
08	40	
09	00	
0A	00	
0B		
0C		
0D		
0E		
0F		
10	00	q
11	00	
12	00	
13	34	
14	4C	
15	4C	
16	34	
17	04	
18	04	
19	00	
1A	00	
1B		
1C		
1D		
1E		
1F		

20	00	r
21	00	
22	00	
23	58	
24	60	
25	40	
26	40	
27	40	
28	00	
29	00	
2A	00	
2B		
2C		
2D		
2E		
2F		
30	00	s
31	00	
32	00	
33	3C	
34	40	
35	38	
36	04	
37	78	
38	00	
39	00	
3A	00	
3B		
3C		
3D		
3E		
3F		

40	00	t
41	20	
42	20	
43	78	
44	20	
45	20	
46	20	
47	18	
48	00	
49	00	
4A	00	
4B		
4C		
4D		
4E		
4F		
50	00	u
51	00	
52	00	
53	44	
54	44	
55	44	
56	44	
57	3C	
58	00	
59	00	
5A	00	
5B		
5C		
5D		
5E		
5F		

60	00	v
61	00	
62	00	
63	44	
64	44	
65	44	
66	28	
67	10	
68	00	
69	00	
6A	00	
6B		
6C		
6D		
6E		
6F		
70	00	w
71	00	
72	00	
73	44	
74	44	
75	44	
76	58	
77	28	
78	00	
79	00	
7A	00	
7B		
7C		
7D		
7E		
7F		